Code No: 123BN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2018
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is the minimal set of connectives required for a well formed formula? [2]
b) Convert the following propositional logic in to DNF $p\Lambda(p\rightarrow q)$. [3]
c) Write the properties of semi group and abelian group. [2]
d) What are the properties of Equivalence relation? [3]
e) How many different outcomes are possible by tossing 20 similar coins? [2]
f) How many integers between 1 and 100 have a sum of digits of integer numbers equal to 10? [3]
g) What is the exponential enumerator for the permutations of 2 objects of one kind and 3 objects of another kind? [2]
h) Find the generating function for the number of ways to distribute 30 balls into 5 numbered boxes where each box contains at least 3 balls and at most 7 balls. [3]
i) Define spanning tree. What are its characteristics? [2]
j) Write the difference between Hamiltonian graphs and Euler graphs. [3]

PART-B

(50 Marks)

2.a) Obtain the principal disjunctive normal form of $p\rightarrow((p\rightarrow q)\Lambda\sim(\sim q\ V\sim p))$.
b) Simplify the given expression by using logical identities $((p\rightarrow q)V(p\rightarrow s))\rightarrow(pVs)$. [5+5]

OR

3.a) Obtain the principal conjunctive normal form of $(p\rightarrow (q\ \Lambda\ r))\ \Lambda\ (\sim p\rightarrow(\sim q\ \Lambda\sim r))$.
b) Using the statements R: Arjun is Happy H: Arjun is rich. Write the following statement in symbolic form,
i) Arjun is neither happy nor rich
ii) Arjun is poor or he is both rich and unhappy. [5+5]

4. Verify which of the following are semi groups.
a) (N,+)
b) (Q,-)
c) (R,+)
d) (Q,.) (Where . (dot) is defined as a.b = a – b + ab). [10]

OR

5.a) If R be a relation in the set of integers z defined by $R= \{(x,y):x\epsilon z,y\epsilon z,(x-y)$ is divisible by $6\}$.
b) If A, B, C are three sets such that $A\subseteq B$, Show $(A\times B)\subseteq (B\times C)$. [5+5]

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6.a) Show that given any 52 integers, there exists two of them whose sum or else whose difference is divisible by 100

b) Prove the identity $1\times1!+2\times2!+3\times3!+\ldots\ldots+n\times n!=(n+1)!-1$. [5+5]

OR

7.a) Using binomial identities evaluate the sum of the following:
1.2.3+2.3.4+……+ (n-2)(n- 1)n.

b) Compute the number of rows of 6 Americans, 7 Mexicans and 10 Canadians in which an American invariably stands between a Mexican and a Canadian and in which a Mexican and a Canadian never stand side by side. [5+5]

8.a) Solve $a_n-5a_{n-1}+6a_{n-2}=2^n+n$, $n\geq2$, given $a_0=1$, $a_1=1$ using generating functions.

b) Find the number of ways in which 4 persons each rolling a single die once, can have a total of 17? [5+5]

OR

9.a) Find the particular solution for the following difference equation:
$a_n+5a_{n-1}+6a_{n-2}=42.4^n$

b) Solve $a_n-2a_{n-1}-3a_{n-2}=5^n$, $n\geq2$, given $a_0=-2$, $a_1=1$. [5+5]

10.a) Let C_n be the cycle graph with n vertices. Prove that C_5 is the only cycle graph isomorphic to its complement.

b) If G is non-directed graph with 12 edges. Suppose that G has 6 vertices of degree 3 and the rest have degree less than 3. Determine the minimum number of vertices, G can have. [5+5]

OR

11. Define following terms:
a) Out degree
b) Simple graph
c) Kuratowski's two graphs. [10]

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